Osmosis in Animals & Plants

Exam code: 0610 & 0970

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Help me understand

Osmosis in Plant Tissues

- When water moves into a plant cell, the vacuole gets bigger, **pushing the cell membrane against the cell wall**
- Water entering the cell by osmosis makes the cell **rigid and firm**
- This is important for plants as the effect of all the cells in a plant being firm is to **provide support and strength for the plant** - making the plant stand upright with its leaves held out to catch sunlight
- The pressure created by the **cell wall** stops too much water entering and **prevents the cell from bursting**
- If plants do not receive enough water the cells cannot remain rigid and firm (turgid) and the plant **wilts**

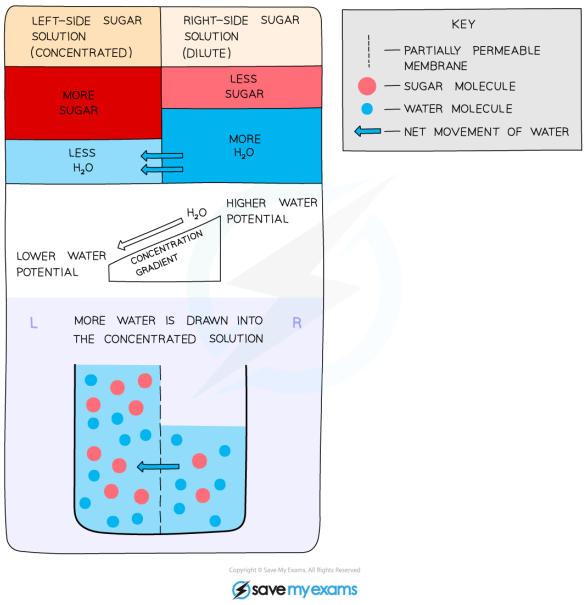
Osmosis: Extended

- Osmosis is the net movement of water molecules from a region of **higher water potential** (dilute solution) to a region of **lower water potential** (concentrated solution), through a **partially permeable membrane**
- **To understand this topic with clarity, it is important to understand the following**
  - High water potential = dilute solution = low solute concentration (right side of diagram)

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



How osmosis works

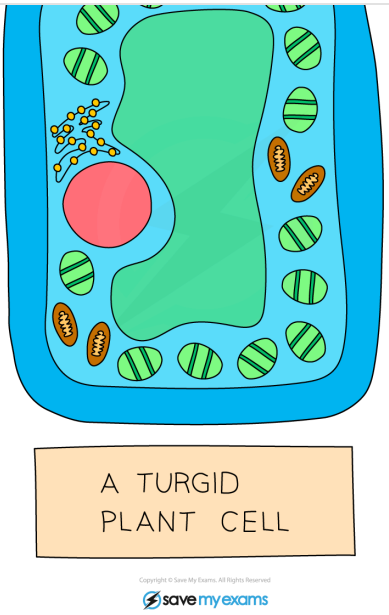
Examiner Tips and Tricks

The best explanations to do with osmosis will refer to **water potential**, so if you are aiming for a 7, 8 or 9 you will need to understand the concept and use it in your explanations.

Osmosis in Animals & Plants: Extended

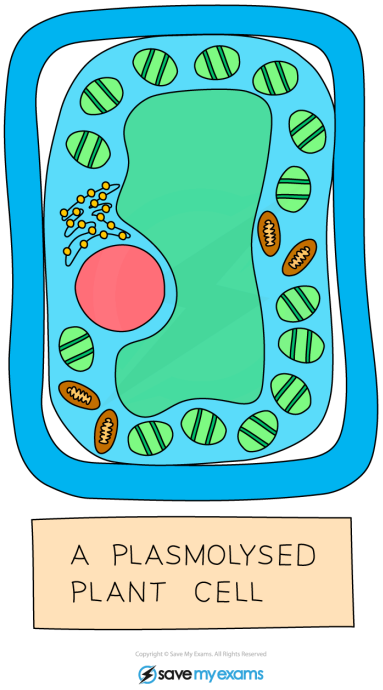
Plant cells in solutions of different concentrations

- When plant cells are placed in a solution that has a **higher water potential** (dilute solution) than inside the cells (e.g. distilled water) then water moves into the plant cells via osmosis
- These water molecules push the cell membrane against the cell wall, increasing the **turgor pressure** in the cells which makes them **turgid**



A turgid plant cell

- When plant cells are placed in a **concentrated** solution (with a lower water potential than inside the cells) water molecules will move **out of the plant cells by osmosis**, making them flaccid
  - If plant cells become flaccid it can negatively affect the plant's ability to support itself
- If looked at underneath the microscope, the plant cells might be **plasmolysed**, meaning the cell membrane has pulled away from the cell wall



A plasmolysed plant cell

- **Animal cells** also lose and gain water as a result of osmosis
- As animal cells **do not have a supporting cell wall**, the results on the cell are more severe
- If an animal cell is placed into a **strong sugar solution** (with a lower water potential than the cell), it will lose water by osmosis and become **crenated** (shrivelled up)
- If an animal cell is placed into **distilled water** (with a higher water potential than the cell), it will gain water by osmosis and, as it has **no cell wall to create turgor pressure**, will continue to do so until the cell membrane is stretched too far and **it bursts**

Effect of osmosis on animal cells

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